CORE PURE / POLAR COORDINATES

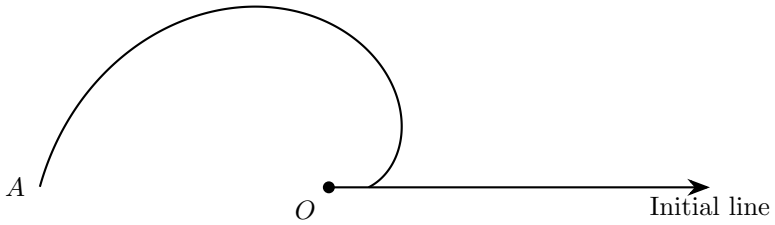
Polar Coordinates Basics

Worked solutions 10 questions · 97 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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1. The polar curves C_1 and C_2 are defined by

$$C_1 : r = 1 + 2\theta, \qquad 0 \leq \theta \leq \pi$$

and

$$C_2 : r = 8 - \theta, \qquad 0 \leq \theta \leq 2\pi$$

The diagram above shows C_1. Its endpoint A lies on the ray opposite the initial line.

(a) State the polar coordinates of A and find its Cartesian coordinates, taking the pole as the origin and the initial line as the positive x-axis. **[3]**

(b) Find the polar coordinates of the point where C_1 and C_2 intersect. **[2]**

(c) On a copy of the diagram above, sketch C_2, indicating the radii of its two endpoints. **[3]**

Solution

(a) The ray opposite the initial line has $\theta = \pi$, giving $r = 1 + 2\pi$. Hence the polar coordinates of A are

$$(1 + 2\pi, \pi)$$

Using $x = r\cos\theta$ and $y = r\sin\theta$:

$$x = (1 + 2\pi)\cos\pi = -1 - 2\pi$$

$$y = (1 + 2\pi)\sin\pi = 0$$

Thus the Cartesian coordinates of A are

$$(-1 - 2\pi, 0)$$

(b) At the intersection, the two radii are equal:

$$1 + 2\theta = 8 - \theta$$

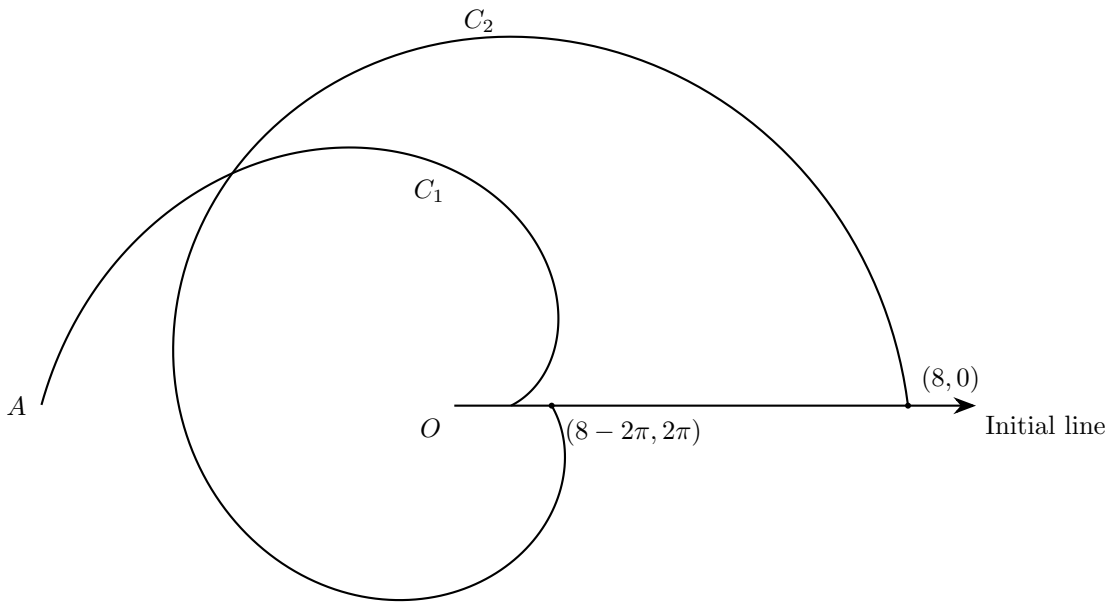
$$3\theta = 7 \quad \Longrightarrow \quad \theta = \frac{7}{3}$$

$$r = 1 + 2\left(\frac{7}{3}\right) = \frac{17}{3}$$

The polar coordinates of the intersection are

$$\left(\frac{17}{3}, \frac{7}{3}\right)$$

(c) The sketch should look like:



As θ increases, C_2 makes one anticlockwise turn inwards, with endpoint radii 8 and $8 - 2\pi$.

2. The curve C has the polar equation

$$r = \frac{6}{2 - \sin\theta}, \qquad -\pi < \theta \leq \pi$$

(a) Verify that the point with polar coordinates $\left(4, \frac{\pi}{6}\right)$ lies on C. **[1]**

(b) Find the exact polar coordinates of the point on C which is furthest from the pole, O. **[3]**

(c) The Cartesian line $y = x$ intersects C at two points. Find the exact Cartesian coordinates of both points. **[4]**

Solution

(a) Substituting $\theta = \frac{\pi}{6}$ into the polar equation gives

$$r = \frac{6}{2 - \sin\frac{\pi}{6}} = \frac{6}{2 - \frac{1}{2}} = 4$$

Thus $\left(4, \frac{\pi}{6}\right)$ lies on C.

(b) Since $-1 \leq \sin\theta \leq 1$, it follows that

$$1 \leq 2 - \sin\theta \leq 3 \quad \Longrightarrow \quad 2 \leq r \leq 6$$

As $r > 0$, the distance from the pole is r. The greatest distance is therefore 6, attained when $\sin\theta = 1$, so $\theta = \frac{\pi}{2}$ in the given interval. The required polar coordinates are

$$\left(6, \frac{\pi}{2}\right)$$

(c) Using $x = r\cos\theta$ and $y = r\sin\theta$, the line $y = x$ gives

$$r\sin\theta = r\cos\theta$$

$$\sin\theta = \cos\theta \qquad (r > 0)$$

$$\tan\theta = 1$$

$$\theta = \frac{\pi}{4} \quad \text{or} \quad -\frac{3\pi}{4}$$

At $\theta = \frac{\pi}{4}$:

$$r = \frac{6}{2 - \frac{\sqrt{2}}{2}} = \frac{12}{4 - \sqrt{2}}$$

$$x = y = \frac{12}{4 - \sqrt{2}} \times \frac{\sqrt{2}}{2} = \frac{6\sqrt{2}(4 + \sqrt{2})}{16 - 2} = \frac{6 + 12\sqrt{2}}{7}$$

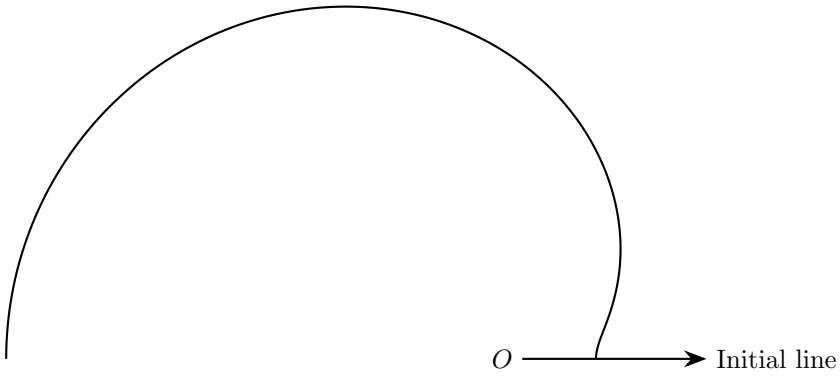
At $\theta = -\frac{3\pi}{4}$:

$$r = \frac{6}{2 + \frac{\sqrt{2}}{2}} = \frac{12}{4 + \sqrt{2}}$$

$$x = y = \frac{12}{4 + \sqrt{2}} \times \left(-\frac{\sqrt{2}}{2} \right) = -\frac{6\sqrt{2}(4 - \sqrt{2})}{16 - 2} = \frac{6 - 12\sqrt{2}}{7}$$

The Cartesian coordinates of the two points are

$$\left(\frac{6 + 12\sqrt{2}}{7}, \frac{6 + 12\sqrt{2}}{7} \right) \quad \text{and} \quad \left(\frac{6 - 12\sqrt{2}}{7}, \frac{6 - 12\sqrt{2}}{7} \right)$$



3. The curve C has polar equation

$$r = 4 - 3\cos\theta, \qquad -\pi < \theta \leq \pi$$

(a) The diagram above shows the portion of C for which $0 \leq \theta \leq \pi$. Complete the sketch of C. **[1]**

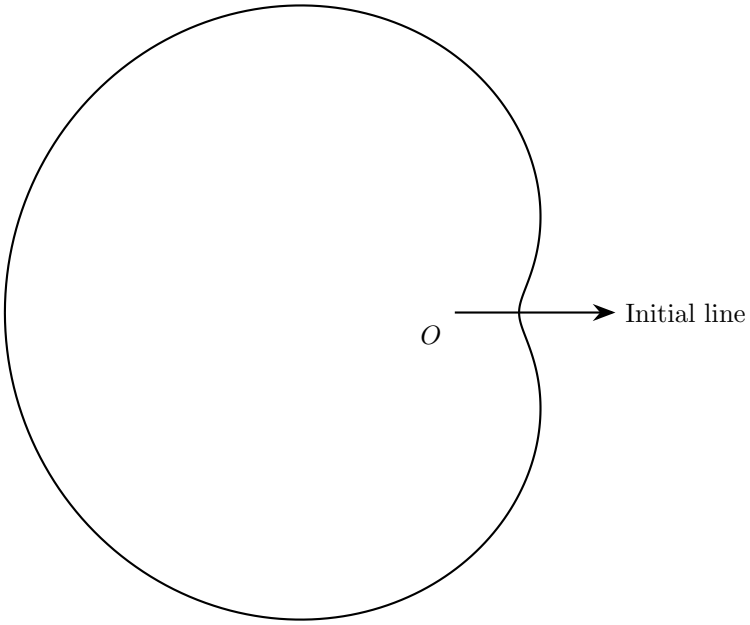
The straight line ℓ has polar equation

$$r\cos\theta = 1$$

(b) Add ℓ to your completed sketch and find all the points of intersection of C and ℓ. Give their polar coordinates, with radii exact and any non-zero angles in radians to 3 significant figures. **[6]**

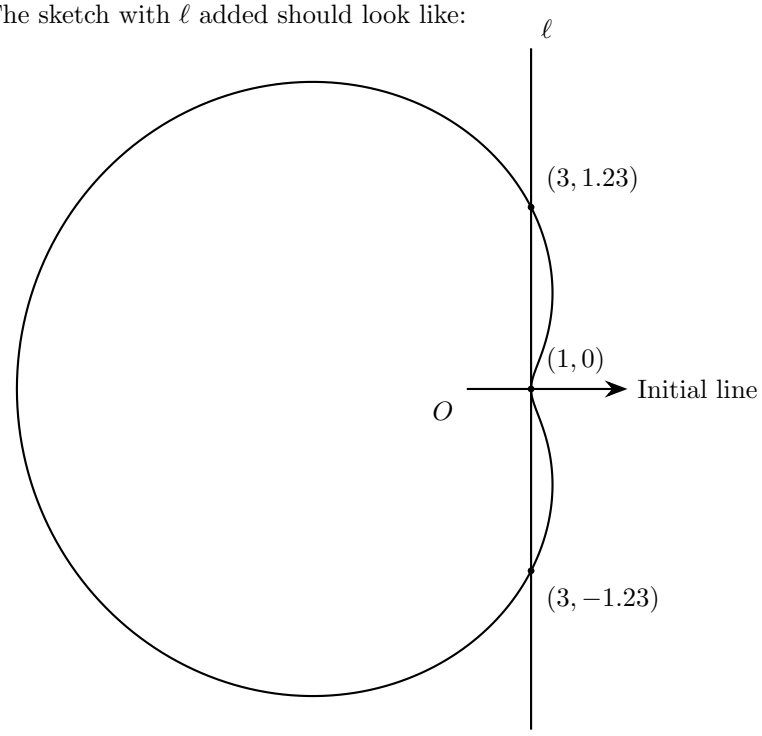
Solution

(a) The completed sketch should look like:



Since $\cos(-\theta) = \cos\theta$, the curve is symmetric about the initial line. Reflect the given portion in the initial line to complete C.

(b) The sketch with ℓ added should look like:



Since $x = r\cos\theta$, the equation of ℓ is $x = 1$, so it is a vertical line through $(1, 0)$.

At an intersection, substitute $r = 4 - 3\cos\theta$ into the equation of ℓ:

$$(4 - 3\cos\theta)\cos\theta = 1$$

$$3\cos^2\theta - 4\cos\theta + 1 = 0$$

$$(3\cos\theta - 1)(\cos\theta - 1) = 0$$

Thus $\cos\theta = 1$ or $\cos\theta = \frac{1}{3}$. In the given range $-\pi < \theta \leq \pi$, these give

$$\cos\theta = 1 \quad \Longrightarrow \quad \theta = 0, \qquad r = 4 - 3 = 1$$

$$\cos\theta = \frac{1}{3} \quad \Longrightarrow \quad \theta = \pm\arccos\left(\frac{1}{3}\right), \qquad r = 4 - 3\left(\frac{1}{3}\right) = 3$$

The polar coordinates are

$$(1, 0), \qquad (3, 1.23), \qquad (3, -1.23) \qquad (3 \text{ s.f.})$$

4. The curve C_1 has polar equation

$$r^2 + 4r\cos\theta - 6r\sin\theta = 12$$

where $r > 0$ and $-\pi < \theta \leq \pi$.

(a) Find a Cartesian equation for C_1 and show that it represents a circle. State its radius and the Cartesian coordinates of its centre. **[4]**

(b) Sketch C_1, marking its centre and the pole. Find the exact greatest and least distances of a point on C_1 from the pole. **[4]**

Solution

(a) Using $x = r\cos\theta$, $y = r\sin\theta$ and $r^2 = x^2 + y^2$, the polar equation becomes

$$x^2 + y^2 + 4x - 6y = 12$$

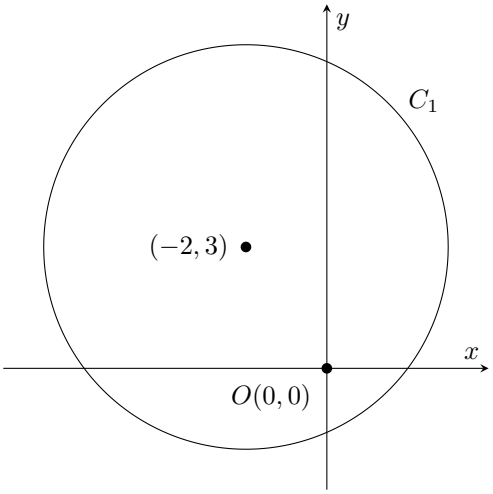
$$(x^2 + 4x + 4) + (y^2 - 6y + 9) = 12 + 4 + 9$$

$$(x + 2)^2 + (y - 3)^2 = 25$$

This has the standard circle form $(x - a)^2 + (y - b)^2 = R^2$, so

$$\text{Centre } (-2, 3), \qquad \text{radius } 5$$

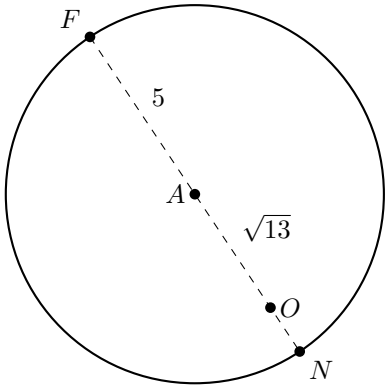
(b) The sketch should look like:



The distance of a point on C_1 from the pole is r, since $r > 0$. The centre A has Cartesian coordinates $(-2, 3)$, so its distance from the pole is

$$OA = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$$

Since $\sqrt{13} < 5$, the pole lies inside the circle. The line through A and O meets C_1 at the furthest point F, beyond A, and at the nearest point N, beyond O:



From the diagram, the extremes are at F and N:

$$\text{Greatest distance} = OF = OA + AF = 5 + \sqrt{13}$$

$$\text{Least distance} = ON = AN - OA = 5 - \sqrt{13}$$

5. **(a)** The curve C_1 has Cartesian equation

$$x^2 + y^2 = \sqrt{3}\, y$$

Find the centre and radius of C_1, and show that its non-pole points have polar equation $r = \sqrt{3}\sin\theta$. **[4]**

The curve C_2 has polar equation

$$r = 1 + \cos\theta$$

Consider both curves for $0 < \theta < \pi$, together with their limiting endpoints.
In this interval, $r > 0$ for both curves. The pole is a limiting endpoint of each curve.

(b) Show that C_1 and C_2 have exactly one intersection other than the pole, and find the exact polar coordinates of this intersection, P. **[4]**

(c) Sketch both curves on a single diagram, clearly labelling each curve. Mark P and indicate the Cartesian coordinates of every point where either curve meets the initial line. **[3]**

Solution

(a) Completing the square:

$$x^2 + y^2 - \sqrt{3}\, y = 0$$

$$x^2 + \left(y - \frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$$

Hence the centre and radius are

$$\text{Centre } \left(0, \frac{\sqrt{3}}{2}\right), \qquad \text{radius } \frac{\sqrt{3}}{2}$$

Substituting $x = r\cos\theta$ and $y = r\sin\theta$ into the original equation:

$$r^2\cos^2\theta + r^2\sin^2\theta = \sqrt{3}\, r\sin\theta$$

$$r^2 = \sqrt{3}\, r\sin\theta$$

At a non-pole point, $r \neq 0$, so division by r gives

$$r = \sqrt{3}\sin\theta$$

(b) Since both radii are positive for $0 < \theta < \pi$, a non-pole intersection must have equal radii at the same angle:

$$\sqrt{3}\sin\theta = 1 + \cos\theta$$

$$\sqrt{3}\sin\theta - \cos\theta = 1$$

Write $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Expanding:

$$R\sin(\theta - \alpha) = R\sin\theta\cos\alpha - R\cos\theta\sin\alpha$$

Comparing the coefficients of $\sin\theta$ and $\cos\theta$:

$$R\cos\alpha = \sqrt{3}, \qquad R\sin\alpha = 1$$

so

$$R = \sqrt{\left(\sqrt{3}\right)^2 + 1^2} = 2, \qquad \tan\alpha = \frac{1}{\sqrt{3}} \quad \Longrightarrow \quad \alpha = \frac{\pi}{6}$$

Hence the equation becomes

$$2\sin\left(\theta - \frac{\pi}{6}\right) = 1$$

$$\sin\left(\theta - \frac{\pi}{6}\right) = \frac{1}{2}$$

For $0 \le \theta \le \pi$, this gives

$$\theta - \frac{\pi}{6} = \frac{\pi}{6} \quad \text{or} \quad \frac{5\pi}{6}, \qquad \text{so} \qquad \theta = \frac{\pi}{3} \quad \text{or} \quad \pi$$

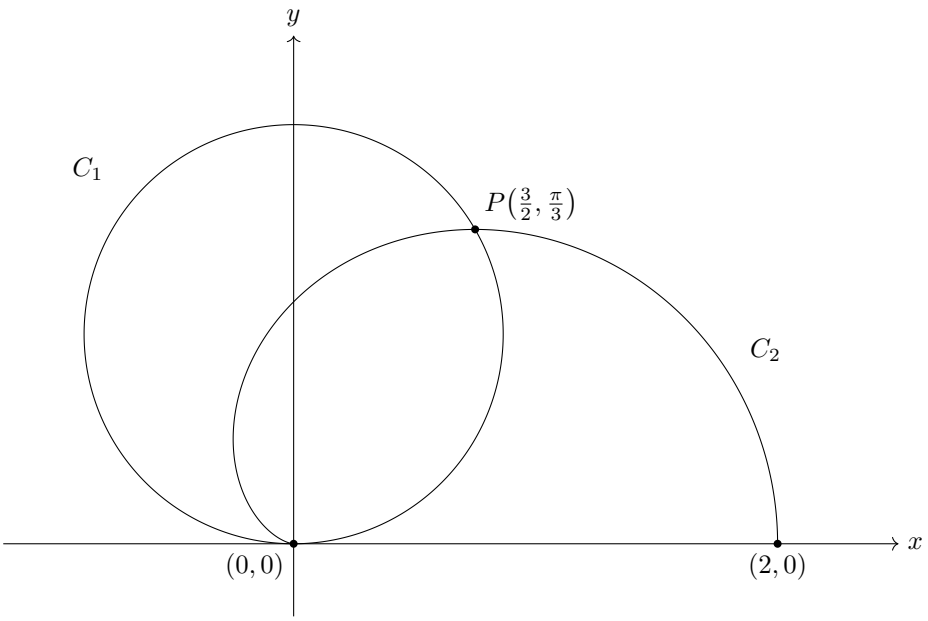
The value $\theta = \pi$ is the pole endpoint and is excluded from the interval. Therefore there is exactly one non-pole intersection. Its radius is

$$r = 1 + \cos\frac{\pi}{3} = \frac{3}{2}$$

so P has polar coordinates

$$\left(\frac{3}{2}, \frac{\pi}{3}\right)$$

(c) The sketch should look like:



C_1 is the full circle from part (a), lying above the initial line and touching it at the pole. At both limiting endpoints, $r = \sqrt{3}\sin\theta = 0$, so its only point on the initial line has Cartesian coordinates $(0, 0)$.

C_2 is the upper half of the cardioid, extending slightly into $x < 0$ before reaching the pole. Since $r(0) = 2$ and $r(\pi) = 0$, its points on the initial line have Cartesian coordinates $(2, 0)$ and $(0, 0)$.

6. A curve has Cartesian equation

$$(x^2 + y^2)^2 = 5x^2 + 4xy + 2y^2$$

where x and y are not both zero. Use the convention $r > 0$.

(i) Show that the polar equation of the curve is

$$r^2 = \frac{7 + 3\cos 2\theta + 4\sin 2\theta}{2}$$ **[4]**

(ii) Hence find the least and greatest distances of points on the curve from the origin, giving your answers in exact form. **[3]**

Solution

(i) Using $x = r\cos\theta$, $y = r\sin\theta$ and $r^2 = x^2 + y^2$, the Cartesian equation becomes

$$r^4 = 5r^2\cos^2\theta + 4r^2\cos\theta\sin\theta + 2r^2\sin^2\theta$$

Since $r > 0$, we can divide by r^2. Applying the double angle formulae gives

$$\begin{aligned} r^2 &= 5\cos^2\theta + 4\cos\theta\sin\theta + 2\sin^2\theta \\ &= \frac{5}{2}(1 + \cos 2\theta) + 2\sin 2\theta + (1 - \cos 2\theta) \\ &= \frac{7}{2} + \frac{3}{2}\cos 2\theta + 2\sin 2\theta \\ &= \frac{7 + 3\cos 2\theta + 4\sin 2\theta}{2} \end{aligned}$$

(ii) Write $3\cos 2\theta + 4\sin 2\theta$ in the form $R\cos(2\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Expanding:

$$R\cos(2\theta - \alpha) = R\cos 2\theta\cos\alpha + R\sin 2\theta\sin\alpha$$

Comparing the coefficients of $\cos 2\theta$ and $\sin 2\theta$:

$$R\cos\alpha = 3, \qquad R\sin\alpha = 4$$

so

$$R = \sqrt{3^2 + 4^2} = 5, \qquad \tan\alpha = \frac{4}{3}$$

Hence, using part (i):

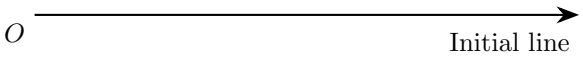
$$r^2 = \frac{7 + 5\cos(2\theta - \alpha)}{2}$$

The cosine takes all values from -1 to 1, so both bounds below are attained:

$$\frac{7 - 5}{2} \le r^2 \le \frac{7 + 5}{2} \quad \Longrightarrow \quad 1 \le r^2 \le 6$$

The distance from the origin is $r > 0$, so

$$\text{Least distance} = 1, \qquad \text{greatest distance} = \sqrt{6}$$



7. The graph G_1 has polar equation

$$r = 6\sec\left(\theta + \frac{\pi}{6}\right), \qquad -\frac{2\pi}{3} < \theta < \frac{\pi}{3}$$

The graph G_2 is the ray with polar equation

$$\theta = -\alpha, \qquad r > 0$$

where $0 < \alpha < \frac{\pi}{2}$.

(a) Show that G_1 is a straight line by finding its Cartesian equation in the form $y = mx + c$. **[3]**

(b) On a copy of the diagram above, sketch G_1 and G_2. Label the angle α and give the polar coordinates of the intersection of G_1 with the initial line. **[3]**

(c) Find, in terms of α, the polar coordinates of the intersection of G_1 and G_2. **[2]**

(d) The region bounded by the initial line, G_1 and G_2 has area $12\sqrt{3}$ square units. Determine the exact value of α. **[4]**

Solution

(a) Using $x = r\cos\theta$ and $y = r\sin\theta$:

$$r = 6\sec\left(\theta + \frac{\pi}{6}\right)$$

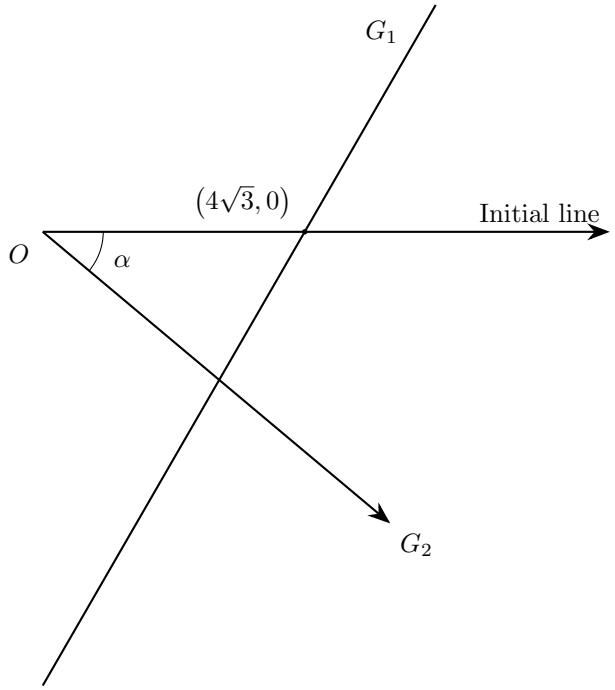
$$r\cos\left(\theta + \frac{\pi}{6}\right) = 6$$

$$r\left(\frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta\right) = 6$$

$$\frac{\sqrt{3}}{2}x - \frac{1}{2}y = 6$$

$$y = \sqrt{3}x - 12 \qquad (m = \sqrt{3},\ c = -12)$$

(b) The sketch should look like:



The ray G_2 makes an angle α clockwise from the initial line. At the intersection of G_1 with the initial line, $\theta = 0$, so the polar coordinates are

$$\left(6\sec\frac{\pi}{6}, 0\right) = \left(4\sqrt{3}, 0\right)$$

(c) At the intersection, $\theta = -\alpha$. Substituting into the equation of G_1:

$$r = 6\sec\left(-\alpha + \frac{\pi}{6}\right)$$

Thus the polar coordinates are

$$\left(6\sec\left(\frac{\pi}{6} - \alpha\right), -\alpha\right)$$

(d) From parts (b) and (c), the two sides of the triangle meeting at O have lengths $4\sqrt{3}$ and $6\sec\left(\frac{\pi}{6} - \alpha\right)$. Their included angle is α, so the triangle area formula gives

$$12\sqrt{3} = \frac{1}{2}\left(4\sqrt{3}\right)\left(6\sec\left(\frac{\pi}{6} - \alpha\right)\right)\sin\alpha$$

$$1 = \frac{\sin\alpha}{\cos\left(\frac{\pi}{6} - \alpha\right)}$$

$$\sin\alpha = \cos\left(\frac{\pi}{6} - \alpha\right)$$

Expanding the cosine:

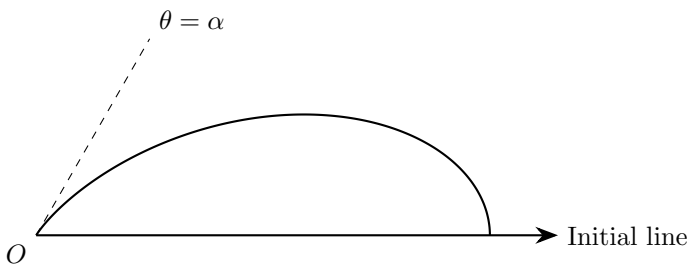
$$\sin\alpha = \frac{\sqrt{3}}{2}\cos\alpha + \frac{1}{2}\sin\alpha$$

$$\frac{1}{2}\sin\alpha = \frac{\sqrt{3}}{2}\cos\alpha$$

$$\tan\alpha = \sqrt{3}$$

Since $0 < \alpha < \frac{\pi}{2}$:

$$\alpha = \frac{\pi}{3}$$



8. A curve has polar equation

$$r = 2(1 + \cos 3\theta), \qquad 0 \leq \theta < 2\pi$$

with only points for which $r > 0$ represented by polar coordinates. The pole O is included as a limiting point of the curve.

(i) The diagram above shows the arc traced from $\theta = 0$ until the curve first reaches O. The tangent there has equation $\theta = \alpha$, where $0 < \alpha < \pi$. Find α in terms of π. **[2]**

(ii) (a) Let $f(\theta) = 2(1 + \cos 3\theta)$. Show that, for every integer k,

$$f\left(\frac{2k\pi}{3} - \theta\right) = f(\theta)$$

and also show that

$$f\left(\theta + \frac{2\pi}{3}\right) = f(\theta)$$

[4]

(b) Hence state the equations of all the distinct lines of symmetry of the curve, using angles in $0 \leq \theta < \pi$. State also the smallest positive angle through which the curve can be rotated about O and remain unchanged. **[3]**

(iii) Sketch the complete curve. State its maximum radius and all the corresponding values of θ in $0 \leq \theta < 2\pi$. **[4]**

Solution

(i) At the pole, the radius tends to zero. The first positive angle at which this happens satisfies

$$2(1 + \cos 3\theta) = 0$$

$$\cos 3\theta = -1$$

$$3\theta = \pi$$

The tangent at the pole is the line through the pole at the angle where $r = 0$, so

$$\alpha = \frac{\pi}{3}$$

(ii) (a) Using the addition formula for cosine, with $\cos 2k\pi = 1$ and $\sin 2k\pi = 0$ for every integer k:

$$\begin{aligned} f\left(\frac{2k\pi}{3} - \theta\right) &= 2[1 + \cos(2k\pi - 3\theta)] \\ &= 2[1 + \cos 2k\pi \cos 3\theta + \sin 2k\pi \sin 3\theta] \\ &= 2(1 + \cos 3\theta) \\ &= f(\theta) \end{aligned}$$

Similarly, using the addition formula for cosine:

$$\begin{aligned} f\left(\theta + \frac{2\pi}{3}\right) &= 2[1 + \cos(3\theta + 2\pi)] \\ &= 2[1 + \cos 3\theta \cos 2\pi - \sin 3\theta \sin 2\pi] \\ &= 2(1 + \cos 3\theta) \\ &= f(\theta) \end{aligned}$$

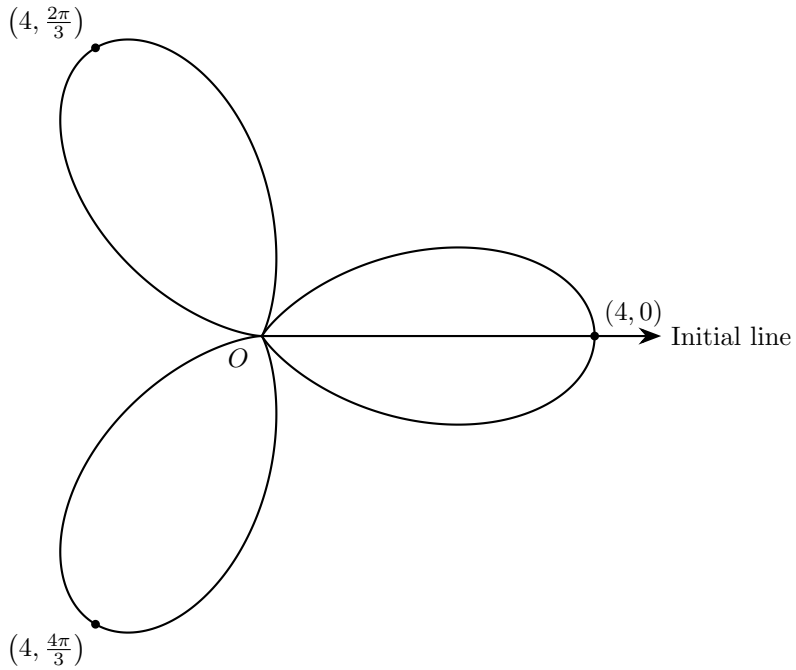
(b) Reflection in a line $\theta = \beta$ replaces θ by $2\beta - \theta$. The first identity in part (ii)(a) therefore gives symmetry lines $\beta = \frac{k\pi}{3}$. The distinct lines in the required range are

$$\theta = 0, \qquad \theta = \frac{\pi}{3}, \qquad \theta = \frac{2\pi}{3}$$

The second identity in part (ii)(a) gives rotational symmetry. Since $\cos 3\theta$ has smallest positive period $\frac{2\pi}{3}$, the smallest positive rotation is

$$\frac{2\pi}{3}$$

(iii) The sketch should look like:



The three identical lobes meet at O and are related by rotations of $\frac{2\pi}{3}$. The labelled points are given in polar coordinates.

Since $\cos 3\theta \leq 1$, the radius satisfies

$$r = 2(1 + \cos 3\theta) \leq 4$$

Equality occurs when

$$\begin{aligned} \cos 3\theta &= 1 \\ 3\theta &= 2k\pi \qquad (k \in \mathbb{Z}) \\ \theta &= \frac{2k\pi}{3} \end{aligned}$$

Thus the maximum radius and all its corresponding angles in $0 \leq \theta < 2\pi$ are

$$r_{\max} = 4, \qquad \theta = 0,\ \frac{2\pi}{3},\ \frac{4\pi}{3}$$

9. **(a)** The line L has polar equation

$$r = 3\sec\left(\theta - \frac{\pi}{4}\right), \qquad -\frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

Show that its Cartesian equation is $x + y = 3\sqrt{2}$. **[2]**

(b) The curve C has polar equation

$$r = 5 - 2\sin 2\theta, \qquad -\pi < \theta \leq \pi$$

Find the exact polar coordinates of all the points of intersection of L and C.
Fully justify that you have found every point of intersection. **[6]**

Solution

(a) Multiplying the polar equation by $\cos\left(\theta - \frac{\pi}{4}\right)$ and expanding:

$$r\cos\left(\theta - \frac{\pi}{4}\right) = 3$$

$$\frac{r\cos\theta + r\sin\theta}{\sqrt{2}} = 3$$

Using $x = r\cos\theta$ and $y = r\sin\theta$, we obtain

$$x + y = 3\sqrt{2}$$

(b) On C, $3 \leq r \leq 7$, so $r > 0$. At an intersection, the equation of L gives

$$\cos\left(\theta - \frac{\pi}{4}\right) = \frac{3}{r}$$

Using the double angle formula:

$$\begin{aligned}\sin 2\theta &= \cos\left(2\theta - \frac{\pi}{2}\right)\\ &= 2\cos^2\left(\theta - \frac{\pi}{4}\right) - 1\\ &= \frac{18}{r^2} - 1\end{aligned}$$

Substituting into the equation of C:

$$r = 5 - 2\left(\frac{18}{r^2} - 1\right)$$

$$r = 7 - \frac{36}{r^2}$$

$$r^3 - 7r^2 + 36 = 0$$

$$(r-3)(r^2 - 4r - 12) = 0$$

$$(r-3)(r-6)(r+2) = 0$$

Since $3 \leq r \leq 7$, the possible radii are $r = 3$ and $r = 6$. The angular restriction on L gives

$$-\frac{\pi}{2} < \theta - \frac{\pi}{4} < \frac{\pi}{2}$$

For $r = 3$:

$$\cos\left(\theta - \frac{\pi}{4}\right) = 1$$
$$\theta - \frac{\pi}{4} = 0$$
$$\theta = \frac{\pi}{4}$$

For $r = 6$:

$$\cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{2}$$
$$\theta - \frac{\pi}{4} = \pm\frac{\pi}{3}$$
$$\theta = -\frac{\pi}{12} \quad \text{or} \quad \frac{7\pi}{12}$$

All transformations above are reversible for $r > 0$, and these are all the solutions of the cosine equations in the allowed interval. All three angles also lie in the domain of C. Hence every intersection has been found, and the exact polar coordinates are

$$\left(3, \frac{\pi}{4}\right), \qquad \left(6, -\frac{\pi}{12}\right), \qquad \left(6, \frac{7\pi}{12}\right)$$

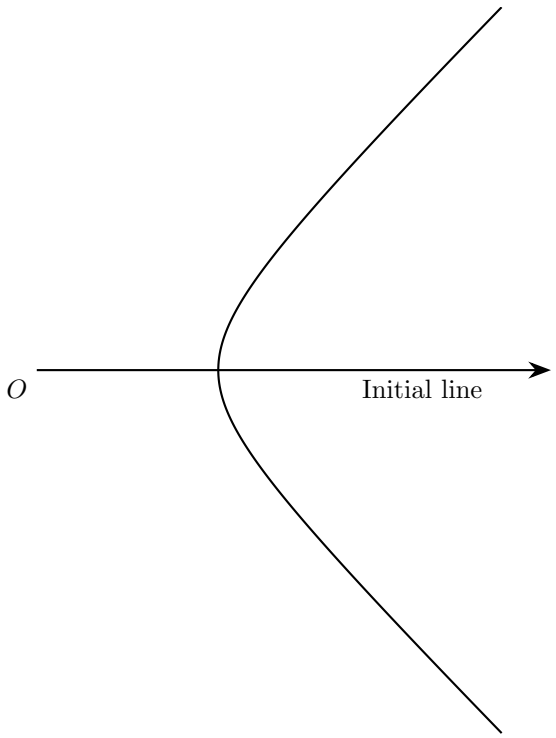
10. **(a)** The curve C is the right-hand branch of the hyperbola

$$x^2 - y^2 = 6x, \qquad x \geq 6$$

Using $x = r\cos\theta$ and $y = r\sin\theta$, show that its polar equation is

$$r = 6\cos\theta\sec 2\theta, \qquad -\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

where $r > 0$. **[3]**



(b) The diagram above shows the curve C.

(i) Find the polar coordinates of the point N on C nearest to the pole O. Justify that your point gives the minimum value of r. **[3]**

(ii) The circle $r = 6\sqrt{3}$ meets C at P and Q, where P lies above the initial line and Q lies below it. Find their exact polar coordinates, using angles in the interval $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. **[5]**

(iii) Find the exact value, in radians, of the angle PNQ. **[4]**

Solution

(a) Substituting $x = r\cos\theta$ and $y = r\sin\theta$:

$$r^2\cos^2\theta - r^2\sin^2\theta = 6r\cos\theta$$

$$r^2\cos 2\theta = 6r\cos\theta$$

$$r\cos 2\theta = 6\cos\theta$$

The last step uses $r > 0$. Since $x > 0$, we may take $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, so $\cos\theta > 0$. Thus

$$\cos 2\theta = \frac{6\cos\theta}{r} > 0$$

Within this interval, $\cos 2\theta > 0$ requires $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Hence the polar equation is

$$r = 6\cos\theta \sec 2\theta, \qquad -\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

(b) (i) On the given interval, $0 < \cos\theta \leq 1$ and $\cos 2\theta > 0$. Therefore

$$\cos 2\theta = \cos^2\theta - \sin^2\theta \leq \cos^2\theta \leq \cos\theta$$

$$r = \frac{6\cos\theta}{\cos 2\theta} \geq 6$$

Equality occurs only when $\theta = 0$, which gives $r = 6$. Thus N has polar coordinates

$$(6, 0)$$

(ii) At an intersection, $r = 6\sqrt{3}$. Using the polar equation from part (a):

$$6\sqrt{3}\cos 2\theta = 6\cos\theta$$

$$\sqrt{3}(2\cos^2\theta - 1) = \cos\theta$$

$$2\sqrt{3}\cos^2\theta - \cos\theta - \sqrt{3} = 0$$

$$(2\cos\theta - \sqrt{3})(\sqrt{3}\cos\theta + 1) = 0$$

$$\cos\theta = \frac{\sqrt{3}}{2} \quad \text{or} \quad -\frac{1}{\sqrt{3}}$$

Since $\cos\theta > 0$ on the given interval, the negative value is excluded. Hence $\theta = \pm\frac{\pi}{6}$. The positive angle gives the point above the initial line, so P has polar coordinates

$$\left(6\sqrt{3}, \frac{\pi}{6}\right)$$

and Q has polar coordinates

$$\left(6\sqrt{3}, -\frac{\pi}{6}\right)$$

(iii) From the polar coordinates, $ON = 6$, $OP = 6\sqrt{3}$ and $\angle NOP = \frac{\pi}{6}$. Applying the cosine rule in triangle ONP:

$$NP^2 = 6^2 + (6\sqrt{3})^2 - 2(6)(6\sqrt{3})\cos\frac{\pi}{6}$$

$$= 36 + 108 - 108 = 36$$

$$NP = 6$$

Since $ON = NP$, triangle ONP is isosceles, with $\angle NPO = \angle NOP = \frac{\pi}{6}$. Therefore

$$\angle ONP = \pi - \frac{\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3}$$

By symmetry about the initial line, $\angle ONQ = \angle ONP$. Using the angles around N:

$$\angle PNQ = 2\pi - 2\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} \text{ radians}$$